FH Innovations Inc. (FHI)

PHASE D

An Enterprise Architecture Proposal

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The Open Group Architecture Framework (TOGAF), 10th Edition

Phase D — Technology Architecture

Infrastructure Transition Plan

Hardware Provisioning — procure baseline office computing hardware (desktops/laptops for Admin) and mobile tablets with mobile data plans for field service staff to log jobs on-site.

Connectivity & Cloud Access — implement reliable office broadband and mobile data for the field to support cloud-based application synchronization, reducing reliance on manual physical handoffs.

Data Security & Backup — establish daily automated backups of financial, billing, and operational records to replace physical index-card reliance and prevent data loss.

Technology Infrastructure Catalog

Component	Type	Purpose
Admin Desktop/Laptop	Physical Technology Component	Office-based data entry, billing, and AR management.
Field Tablet / Ruggedized Device	Physical Technology Component	On-site Job Order logging by service technicians.
Cloud Inventory & Invoicing SaaS	System Software	Central application hosting Inventory, Job Order, and Billing/AR modules.
Automated Backup Service	System Software	Daily backup of financial and operational records.

Network & Data Topology

Field tablets synchronize Job Order and inventory data to the cloud over mobile data or Wi-Fi. The administrative dashboard consumes this synchronized data to drive billing and AR. Automated backups run independently of the operational sync path.

Gap Analysis and Risk Mitigation

Identified Operational Gap	Architectural Impact	Proposed Remediation Strategy
1. No reconciliation between items bought and delivered	Inventory leakage, financial discrepancies, unaccounted stock.	Implement a digital 3-way match protocol: Purchase Order vs. Delivery Receipt vs. Consignment Usage Report.
2. Distance problem with Balintawak supplier causing stockouts	Operational downtime, staff forced to scramble for stock.	Implement safety-stock thresholds with automated lead-time buffers and scheduled bulk ordering.
3. Unsystematic AR aging and long outstanding payments	Cash-flow compression, delayed follow-ups, manual statement creation.	Introduce automated AR aging dashboards with rule-based triggers for follow-up letter generation and structured collection days.
4. Manual tracking via index cards and unverified pre-numbered forms	Human error, missing job orders, inability to track cancellations.	Enforce mandatory digital validation of pre-numbered delivery receipts and Job Orders prior to closing out service tickets.

Per-domain baseline to target gaps

Domain	Baseline	Target	Gap / candidate work package
Business	Paper, ad hoc processes	Defined digital processes with roles and KPIs	Process redesign, SOPs, training (WP-1)
Data	Cards and spreadsheets, no master data	Five centralized entities	Data cleanup and migration of open dealers, items, AR (WP-2)
Application	None	Three-module SaaS	SaaS selection and configuration (WP-3)
Technology	No infrastructure	Laptops, tablets, mobile data, backup	Procurement and setup (WP-4)

Roadmap (Phases E/F Summary)

Wave	Content	Outcome
1. Foundation	Sponsor sign-off; SaaS selection (WP-3); admin hardware and backup (WP-4); master data setup (WP-2)	Administrative data logging in place
2. Pilot	Digital JO workflow and consignment ledger at one dealer, e.g., Toyota Balintawak; tablets and offline sync tested (WP-1, WP-4)	Validated process and connectivity
3. Rollout	Remaining dealers; three-way match and AR aging automation live	Full traceability and automated follow-ups
4. Optimize	KPI review, reorder tuning, dashboard refinement	Targets in Section 3.3 tracked

Traceability Matrix

Driver	Goal	Principle	Requirement	Gap	Component	KPI
Revenue collection and cash flow	Working capital optimization	P3	FR-07, FR-08	Gap 3	Billing & AR Module; AR Ledger	DSO; 61+ day AR share
Availability of supplies	Supply chain resilience	P1	FR-05	Gap 2	Inventory Module; Inventory Master	Stockout incidents
Partner dealer relationships	Inventory and consignment reconciliation	P2	FR-03, FR-04, FR-06	Gap 1	Inventory and Billing modules; Consignment & Billing DB	Reconciliation variance; billing cycle time
Reliable service delivery	Process automation	P1, P2	FR-01, FR-02, NFR-02	Gap 4	Service & JO Module; JO Ledger; field tablets	JO digital validation rate
Ability to scale	Process automation	P1	NFR-05, NFR-06	All	Cloud SaaS	All

Architecture Building Blocks

ABB	Description	Candidate SBB (decided later)
Inventory and consignment tracking	Stock, PO, delivery receipt, reorder alerts	Cloud inventory SaaS
Digital Job Order	Pre-numbered, offline-capable job logging	SaaS job/service module
Billing and AR	Statements, aging, bank reconciliation	SaaS invoicing module
Field mobility	Rugged tablets with mobile data	Commercial tablets and data plans
Backup and security	Daily backup, role-based access, MFA	Vendor or third-party backup

Conclusion and Next Steps

This unified architecture takes FHI from a manual, high-friction operating model to a digitized, traceable one, using a metamodel and content framework deliberately tailored to a flat, 27-person MSME rather than a full enterprise-grade deployment. The three architecture principles — Digital-First Over Manual Tracking, Three-Way Consignment Reconciliation, and Automated AR Aging — give a small, enforceable set of rules that resolve the four operational gaps identified without requiring a dedicated IT department.

Recommended next steps are to secure sponsor sign-off on the Request for Architecture Work, baseline KPIs, verify connectivity at each dealer bay, select a lightweight cloud-based Inventory & Invoicing SaaS consistent with the Application Architecture, and pilot the digital Job Order workflow at a single dealer partner (e.g., Toyota Balintawak) before rolling out across all partner dealerships — consistent with the phased, non-disruptive Operational Integration approach defined in the Framework Interface Assessment.